Patient Health Risk Prediction System

Project Documentation: libraries, process, hyperparameters and results

1. Project overview

Goal: predict whether a patient has a disease (**Has_Disease**, 0/1) from patient health records, so high-risk patients can be flagged early. The project follows the CRISP-ML(Q) methodology: business and data understanding, data preparation, feature engineering, model engineering, evaluation, deployment, monitoring.

Business, economic and success criteria (from the notebook)

Item	Description
Business problem	Hospitals need to identify patients who are at high health risk so that doctors can give them earlier attention and treatment.
Economic problem	Early identification of high-risk patients can help reduce treatment costs, emergency expenses, and unnecessary hospital resources.
ML success criteria	Recall ≥ 80%: identify at least 80% of high-risk patients. F1-score ≥ 80%: keep a good balance between precision and recall. ROC-AUC ≥ 0.80: effectively distinguish between high-risk and low-risk patients.
Economic success criteria	Reduce healthcare costs: reduce unnecessary treatment and emergency-care expenses by identifying high-risk patients early. Reduce resource usage: help hospitals use doctors, beds and medical resources more efficiently. Reduce complications: reduce costly health complications through early identification and intervention.

Item	Value
Raw dataset	Patient_Health_Records_Raw.csv, 10,000 rows × 17 columns
Rows after cleaning	4,971 (rows with unknown / missing target removed, duplicates removed)
Target	Has_Disease (binary classification)
Train / test split	80% / 20% (3,976 / 995 rows), random_state=0
Class balance (train)	Class 0: 2,028 (51.0%) · Class 1: 1,948 (49.0%)
Features after encoding	18 in the original run (more after adding missing flags); 6 kept by SelectKBest

2. Libraries used and their purpose

Library / module	Used for
pandas, numpy	Loading the CSV, cleaning, type conversion, feature creation, array operations
matplotlib, seaborn	Histograms, box plots, count plots, scatter plots, heatmaps (EDA)
scipy.stats (probplot, chi2_contingency)	QQ plots for normality, chi-square tests for categorical vs target
sklearn.model_selection	train_test_split (data split), GridSearchCV (hyperparameter tuning with 5-fold CV)
sklearn.impute.SimpleImputer	Missing-value imputation (median for numeric, most frequent for categorical)
sklearn.preprocessing	OneHotEncoder, OrdinalEncoder (encoding); StandardScaler (scaling)
sklearn.feature_selection	SelectKBest with f_classif (feature selection); mutual_info_classif (check)
sklearn.linear_model / tree / ensemble / neighbors / svm / naive_bayes	LogisticRegression, DecisionTreeClassifier, RandomForestClassifier, KNeighborsClassifier, SVC, GaussianNB
sklearn.metrics	accuracy_score, confusion_matrix, classification_report, f1_score, roc_auc_score
pickle	Saving fitted imputers, encoders, selector, scaler and the model to disk, and loading them for prediction
imblearn (SMOTE)	Imported for class balancing; not applied because classes are already balanced
warnings	Suppress warning messages

3. Process, step by step

3.1 Data cleaning

Column	Cleaning applied
Patient_ID, Name	Dropped (identifiers, not predictive)
Has_Disease (target)	Rows with “unknown”/missing target removed; converted to integer
Age	Converted to numeric; values <0 or >120 set to missing
Gender	Lower-cased and mapped to Male / Female; “unknown” → missing
City	Spelling variants merged (e.g. “la”, “newyork”, “Nwe Yrok”); later dropped from the model
BMI	“kg/m2” text removed, converted to numeric
Blood_Pressure	Formats like “120 over 80” and “120 - 80” unified, split into Systolic_BP and Diastolic_BP
Heart_Rate	Numeric; values ≤0 or >200 set to missing
Cholesterol_Level	Split into Cholesterol_Value (numeric) and Cholesterol_Status (normal / high)
Diabetic, Smoker	Standardized to Yes / No (Smoker also Former)
Medications	Separators standardized (;), “none” → missing, title-cased
Last_Visit_Date	Mixed formats parsed to datetime (dayfirst); Visit_Year extracted
Follow_Up	“two weeks” → 14, converted to numeric days
Duplicates	Duplicate rows removed

3.2 Exploratory data analysis

Descriptive statistics (mean, median, mode, quartiles, IQR, skewness, kurtosis, variance), histograms, box plots, count plots, QQ plots, categorical-vs-target and numeric-vs-target plots, chi-square tests, a correlation analysis and multivariate analysis. Correlations of numeric features with the target are very weak (all within about ±0.04), and gender, diabetic and smoker split almost 50/50 by target.

3.3 Train/test split, missing-value flags and imputation

Order: split first, then add flags, then impute. The split is done **before** imputation to avoid data leakage. A **was_missing** flag (1 = value was blank, 0 = real reading) is added for each column **before** imputing, so the model can tell real readings from filled-in medians. After imputation `isna()` would return all zeros, so the flag cell must run first.

Column	Missing in train (of 3,976)	%
Visit_Year	3,003	76%
Age	2,953	74%
Heart_Rate	2,951	74%
Cholesterol_Status	2,418	61%
Cholesterol_Value	2,385	60%
Follow_Up	2,057	52%
BMI	1,932	49%
Medications	1,620	41%
Gender / Diabetic / Smoker	1,281 – 1,341	32–34%
Systolic_BP / Diastolic_BP	824	21%

Diastolic_BP and Cholesterol_Status flags duplicate the Systolic_BP and Cholesterol_Value flags (same source column), so they are dropped; 11 flags remain. Imputers are fitted on training data only and then applied to the test data.

The split is done **before** imputation to avoid data leakage. Imputers are fitted on training data only and then applied to the test data.

Step	Library / class	Setting
Split	train_test_split	test_size=0.20, random_state=0
Numeric imputation	SimpleImputer	strategy='median' (Age 50, BMI 22.5, Heart_Rate 86, Follow_Up 14, Cholesterol_Value 190, Systolic_BP 120, Diastolic_BP 80)
Categorical imputation	SimpleImputer	strategy='most_frequent'
Visit_Year imputation	SimpleImputer	strategy='median' (2022)

3.4 Feature engineering

New feature	How it is built
Medication_Count	Number of medicines in the Medications text
Age_Group	pd.cut on Age, bins [0,18,40,60,100] → Young, Adult, Middle_Age, Senior
BMI_Category	pd.cut on BMI, bins [0,18.5,25,30,100] → Underweight, Normal, Overweight, Obese
BP_Category	Normal / Elevated / Hypertensive from systolic and diastolic values (np.select)
Risk_Points, Risk_Score	Points for age, BMI, BP, heart rate, cholesterol, diabetes and smoking; bins [-1,2,5,100] → Low, Medium, High

3.5 Encoding

Encoder	Columns	Hyperparameters
OneHotEncoder	Gender, Diabetic, Smoker, Medications, Cholesterol_Status	handle_unknown='ignore', drop='first', sparse_output=False
OrdinalEncoder	Age_Group, BMI_Category, Risk_Score	categories in explicit order (Young→Senior, Underweight→Obese, Low→High), handle_unknown='use_encoded_value', unknown_value=-1

Numeric columns kept as-is: Age, BMI, Heart_Rate, Follow_Up, Cholesterol_Value, Systolic_BP, Diastolic_BP, Visit_Year, Medication_Count, plus the *_was_missing flags. The original run (without flags) gave 3,976 × 18 (train) and 995 × 18 (test) with no missing or infinite values; with the flags there are 11 more columns.

3.6 Feature selection and scaling

Step	Library / class	Setting
Feature selection	SelectKBest	score_func=f_classif, k=6
Scaling	StandardScaler	default; fitted on the 6 selected training features only

Selected features: **Age, Heart_Rate, Follow_Up, Cholesterol_Status_normal, Age_Group, BMI_Category**.

Saved files (pickle): num_imputer.pkl, cat_imputer.pkl, year_imputer.pkl, onehot_encoder.pkl, ordinal_encoder.pkl, SelectKBest_selector.pkl, SelectKBest_scaler.pkl, RandomForest_model.pkl, plus encoded_columns.pkl and flag_cols.pkl (needed so deployment rebuilds columns in the same order). Every file must be re-saved from the same run after adding the flags.

4. Models and hyperparameters

4.1 Baseline models (before tuning)

All seven models were first trained on the 6 scaled features with fixed settings and compared on test accuracy.

Model	Fixed hyperparameters	Test accuracy
Logistic Regression	max_iter=1000, random_state=0	0.4824
SVM (linear)	kernel='linear', random_state=0	0.4693
SVM (RBF)	kernel='rbf', random_state=0	0.4744
KNN	n_neighbors=6, metric='minkowski', p=2	0.5317
Naive Bayes	GaussianNB defaults	0.4854
Decision Tree	criterion='entropy', random_state=0	0.5005
Random Forest	n_estimators=10, criterion='entropy', random_state=0	0.4935

4.2 Hyperparameter tuning with GridSearchCV

Every model was tuned with **GridSearchCV** using 5-fold cross-validation, scoring='f1_weighted', refit=True, n_jobs=-1, verbose=3, on the scaled training data.

Model	Search grid	Candidates / fits	Best hyperparameters
KNN	n_neighbors [3,5,7,9]; weights [uniform, distance]; metric [euclidean, manhattan, minkowski]; p [1,2]	48 / 240	metric='manhattan', n_neighbors=9, p=1, weights='uniform' (best CV F1 0.5005)
Decision Tree (random_state=42)	criterion [gini, entropy]; max_features [None, sqrt, log2]; splitter [best, random]	12 / 60	criterion='gini', max_features='sqrt', splitter='best'
Logistic Regression (max_iter=5000, random_state=42)	solver [newton-cg, lbfgs, liblinear, saga]; penalty [l2]	4 / 20	penalty='l2', solver='newton-cg'
Naive Bayes	var_smoothing [1e-9, 1e-8, 1e-7]	3 / 15	var_smoothing=1e-09
Random Forest (random_state=42)	criterion [gini, entropy]; max_features [log2, sqrt, None]; n_estimators [10, 50, 100]	18 / 90	criterion='gini', max_features=None, n_estimators=10
SVM (SVC, probability=True, random_state=42)	kernel [linear, rbf, poly, sigmoid]; gamma [auto, scale]; C [10, 100, 1000, 2000, 3000]	40 / 200	Not run (code is commented out in the notebook)

5. Evaluation results (test set, 995 rows)

These scores come from the original run, before the missing-value flags were added. Re-run the notebook and update this table after the flags are in.

Tuned model	Accuracy	Weighted F1	ROC-AUC	Class-1 recall	Confusion matrix [[TN FP][FN TP]]
KNN	0.52	0.4985	0.5147	0.73	[[146 338][138 373]]
Decision Tree	0.49	0.4562	0.5214	0.24	[[365 119][387 124]]
Logistic Regression	0.48	0.4742	0.4872	0.36	[[296 188][327 184]]
Naive Bayes	0.49	0.4810	0.4863	0.40	[[281 203][309 202]]
Random Forest	0.51	0.4976	0.5122	0.38	[[312 172][319 192]]

Reading the results. All tuned models score about 0.49–0.52 accuracy and ROC-AUC of about 0.49–0.52, which is close to random guessing on a balanced target. None of them meet the stated success criteria (Recall, F1 ≥ 80%, ROC-AUC ≥ 0.80). This agrees with the EDA, where no feature has a meaningful relationship with Has_Disease (correlations ≈ 0, near-identical class splits). The limiting factor is the data signal, not the choice of model or hyperparameters; the dataset looks synthetic or the target is not driven by these columns.

6. Deployment

A new patient goes through exactly the same steps as training. Raw fields (use None for unknown values) are processed by a single function `predict_patient()`:

Step	Action	Saved object used
1	Add *_was_missing flags (before imputing)	flag_cols.pkl
2	Impute numeric (median), Visit_Year (median), categorical (most frequent)	num_imputer, year_imputer, cat_imputer
3	Feature engineering: Medication_Count, Age_Group, BMI_Category, Risk_Points / Risk_Score	same functions as training
4	One-hot and ordinal encoding, columns put in training order	onehot_encoder, ordinal_encoder, encoded_columns
5	SelectKBest transform, StandardScaler transform	SelectKBest_selector, SelectKBest_scaler
6	predict() and predict_proba()	RandomForest_model.pkl

Notes to check:

• Save grid.best_estimator_ as the final model; the old deployment cell saved the baseline Random Forest from the loop (n_estimators=10, entropy).

• Medication_Count is built after imputation in the notebook, so every row gets the same filled medication string and the value is constant (2). Building it before the imputer step would make it informative.

• Choose the final model by cross-validation F1, not by the test score.

• Earlier results were near random (accuracy and ROC-AUC about 0.5); the flags may help only slightly, because missingness is the only new signal.

Part B. Detailed analysis summary (from the project notebook)

All numbers below are copied from the notebook outputs (cleaned data, 4,971 rows). Statistics are computed on the values that were actually present; many columns have blanks (see section 10), so the count behind each statistic is smaller than 4,971.

7. Univariate statistics of numerical variables (full table)

Statistic	Age	BMI	Heart_Rate	Follow_Up	Has_Disease	Chol_Value	Systolic_BP	Diastolic_BP
mean	49.138	23.593	85.040	21.960	0.495	219.234	121.228	80.029
median	50.0	22.5	85.0	14.0	0.0	190.0	120.0	80.0
mode	6	22.5	63	14	0	190	120	80
Q1 (25%)	23.0	22.5	67.0	14.0	0.0	190.0	120.0	80.0
Q3 (75%)	75.0	24.5	103.0	30.0	1.0	250.0	120.0	80.0
99th pct	99.0	34.6	120.0	30.0	1.0	250.0	157.0	99.0
IQR	52.0	2.0	36.0	16.0	1.0	60.0	0.0	0.0
1.5×IQR	78.0	3.0	54.0	24.0	1.5	90.0	0.0	0.0
Lower fence	-55.0	19.5	13.0	-10.0	-1.5	100.0	120.0	80.0
Upper fence	153.0	27.5	157.0	54.0	2.5	340.0	120.0	80.0
Min	0	15.0	50	14	0	190	90	60
Max	100	35.0	120	30	1	250	160	100
Skewness	0.026	0.879	-0.012	0.010	0.021	0.051	1.039	-0.058
Kurtosis	-1.248	0.734	-1.187	-2.002	-2.000	-1.999	4.255	3.955
Variance	865.003	17.254	420.059	64.025	0.250	899.864	111.099	36.927
Std. dev.	29.411	4.154	20.495	8.002	0.500	29.998	10.540	6.077

Summary of each variable

Variable	Summary
Age	Mean 49.1 ≈ median 50; range 0–100; skew ≈ 0, kurtosis –1.25 → almost uniform, flat distribution with no outliers (fences –55 to 153).
BMI	Median 22.5 equals Q1 and mode: a large block of identical values (22.5) makes IQR only 2; skew +0.88 (right tail up to 35) and kurtosis 0.73; values above 27.5 or below 19.5 are flagged as outliers by the 1.5×IQR rule.
Heart_Rate	Mean 85 ≈ median 85, range 50–120, skew ≈ 0, kurtosis –1.19 → flat, uniform-like; no outliers.
Follow_Up	Only two values, 14 and 30 days (median 14, mean 22); kurtosis –2.0 → two-point distribution, not a continuous variable.
Has_Disease	Target; mean 0.495 → 49.5% positive; kurtosis –2.0 (two classes), balanced.
Cholesterol_Value	Only two values, 190 and 250 (mean 219); skew ≈ 0, kurtosis –2.0 → two-point variable.
Systolic_BP	Median = mode = Q1 = Q3 = 120 (IQR 0); skew +1.04, kurtosis 4.26 → a very sharp spike at 120 with a thin tail from 90 to 160; the box plot flags every other value as an outlier, which is not meaningful.
Diastolic_BP	Median = mode = 80 (IQR 0); skew ≈ –0.06, kurtosis 3.95 → sharp spike at 80 with thin tails 60–100 (same IQR caveat).

Value counts, one line per feature

Feature	One-line value-count summary (non-missing values)
Gender	Female 1,658 (50.5%), Male 1,628 (49.5%); 1,685 missing.
City	New York 1,990 (44.9%), Chicago 1,020 (23.0%), Los Angeles 929 (21.0%), Boston 492 (11.1%); 540 missing.
Diabetic	No 1,710 (51.5%), Yes 1,612 (48.5%); 1,649 missing.
Smoker	Former 1,710 (50.7%), No 837 (24.8%), Yes 827 (24.5%); 1,597 missing.
Medications	Aspirin; Metformin 1,970 (66.0%), Lisinopril; Metformin 1,015 (34.0%); 1,986 missing.
Diagnosis_Code	B20 880, E11.9 813, I10 811, A00 777 (all about 24–27%); 1,690 missing.
Notes	“BP high, needs recheck” 1,018, “Follow up soon” 992, “Patient feeling fine” 917 (about 31–35% each); 2,044 missing.
Cholesterol_Level (raw)	190: 1,024; normal: 1,015; 250: 973; high: 928 (about 24–26% each); 1,031 missing.
Cholesterol_Status	normal 1,015 (52.2%), high 928 (47.8%); 3,028 missing.
Cholesterol_Value	190: 1,024, 250: 973 (two values only); 2,974 missing.
Systolic_BP	120 occurs 2,960 times (about 75% of non-missing); the other 70 values appear only 8–27 times each.
Diastolic_BP	80 occurs 2,974 times (about 75% of non-missing); other values appear only about 22–37 times each.
Has_Disease (target)	0: 2,512 (50.5%), 1: 2,459 (49.5%) → balanced, no resampling needed.
Last_Visit_Date	Range 2020-01-01 to 2025-10-01; visits spread fairly evenly across years (about 45–85 per histogram bin).

8. Four moments: business decision summary (before and after preprocessing)

Before preprocessing = values measured on the cleaned data (table above). After preprocessing = the effect of median imputation, 1%/99% percentile capping (all numeric columns except Systolic_BP and Diastolic_BP) and Yeo-Johnson transformation. The exact after-values are not quoted here because they come from re-running the final notebook, which prints a before/after skewness and kurtosis table; the expected direction is described instead.

Moment	Measure	Before preprocessing (measured)	Business / health decision	After preprocessing (expected effect)
1st – Mean / Median / Mode	Centre	Age 49.1 / 50 / 6; BMI 23.6 / 22.5 / 22.5; Heart rate 85.0 / 85 / 63; Follow-up 22.0 / 14 / 14; Cholesterol 219 / 190 / 190; Systolic 121.2 / 120 / 120; Diastolic 80.0 / 80 / 80	Typical patient is middle-aged, normal BMI, normal BP (120/80), cholesterol at the lower level; mean > median for BMI, follow-up, cholesterol, systolic shows a minority of higher values. Use the median as the typical value because many values are identical.	Missing values are filled with the training median, so median stays the same and more rows sit exactly at the median (Age 50, BMI 22.5, Heart_Rate 86, Follow_Up 14, Cholesterol 190, Systolic 120, Diastolic 80). Capping moves the mean only slightly.
2nd – Variance / Std / Range	Spread	Std: Age 29.4, BMI 4.15, Heart rate 20.5, Follow-up 8.0, Cholesterol 30.0, Systolic 10.5, Diastolic 6.1; ranges 100, 20, 70, 16, 60, 70, 40	Age and heart rate vary widely, so risk grouping (age bands, heart-rate bands) is useful; BP varies little, so only the few extreme readings identify hypertensive patients.	Imputing the median shrinks the variance (more identical values); capping trims the extremes (BMI 15–35 → about 1st–99th percentile); Yeo-Johnson (standardize=False) changes the scale of the values; StandardScaler later gives every selected feature mean 0 and std 1.

Moment	Measure	Before preprocessing (measured)	Business / health decision	After preprocessing (expected effect)
3rd – Skewness	Symmetry	Age 0.03, Heart rate –0.01, Follow-up 0.01, Cholesterol 0.05, Diastolic –0.06 (symmetric); BMI 0.88 and Systolic 1.04 (right-skewed)	Symmetric variables can be used as they are. For BMI and Systolic the tail represents obese / hypertensive patients, who are exactly the high-risk group, so tails are kept (BP is deliberately not capped).	Yeo-Johnson should pull BMI and the other skewed columns toward 0 skew; because of the spike at the median they cannot become perfectly normal.
4th – Kurtosis	Tails / peakedness	Age –1.25, Heart rate –1.19 (flat); Follow-up –2.0, Cholesterol –2.0, Has_Disease –2.0 (two-point); BMI 0.73; Systolic 4.26, Diastolic 3.95 (sharp peak with tails)	Flat variables give no natural cut-off, so clinical bins are needed; the high kurtosis of BP means most patients are normal and a small number are far from normal and need attention.	Capping lowers the kurtosis of BMI (extreme 1% removed); the spike remains, so QQ plots still bend at the ends and kurtosis stays away from 0; models used (tree-based, scaled KNN/logistic) do not require normality.

9. Frequency / count and percentage analysis of categorical variables

Variable	Categories (count, %)	Mode	Unique	Reading
Gender	Female 1,658 (50.5%); Male 1,628 (49.5%)	Female	2	Balanced
City	New York 1,990 (44.9%); Chicago 1,020 (23.0%); Los Angeles 929 (21.0%); Boston 492 (11.1%)	New York	4	New York dominates
Cholesterol_Level	190: 1,024 (26.0%); normal: 1,015 (25.8%); 250: 973 (24.7%); high: 928 (23.6%)	190	4	Mixed numeric/text codes; split into Value + Status
Diabetic	No 1,710 (51.5%); Yes 1,612 (48.5%)	No	2	Balanced
Smoker	Former 1,710 (50.7%); No 837 (24.8%); Yes 827 (24.5%)	Former	3	Half are ex-smokers
Medications	Aspirin; Metformin 1,970 (66.0%); Lisinopril; Metformin 1,015 (34.0%)	Aspirin; Metformin	2	Metformin in every prescription
Diagnosis_Code	B20 880 (26.8%); E11.9 813 (24.8%); I10 811 (24.7%); A00 777 (23.7%)	B20	4	Even spread
Notes	BP high, needs recheck 1,018 (34.8%); Follow up soon 992 (33.9%); Patient feeling fine 917 (31.3%)	BP high…	3	Even spread
Cholesterol_Status	normal 1,015 (52.2%); high 928 (47.8%)	normal	2	Balanced

Percentages are of non-missing values. No category is rare; all categorical variables are either balanced or have one moderately dominant class.

10. Missing-value summary

Counts on the full cleaned data (4,971 rows) after cleaning invalid values (e.g. Age <0 or >120, Heart_Rate ≤0 or >200) to missing. Only the target has no missing values.

Column	Missing	%	Column	Missing	%
Last_Visit_Date	3,759	75.6%	Notes	2,044	41.1%
Age	3,699	74.4%	Medications	1,986	40.0%
Heart_Rate	3,689	74.2%	Diagnosis_Code	1,690	34.0%
Cholesterol_Status	3,028	60.9%	Gender	1,685	33.9%

Column	Missing	%	Column	Missing	%
Cholesterol_Value	2,974	59.8%	Diabetic	1,649	33.2%
Follow_Up	2,547	51.2%	Smoker	1,597	32.1%
BMI	2,426	48.8%	Blood_Pressure / Systolic / Diastolic	1,032	20.8%
Cholesterol_Level	1,031	20.7%	City	540	10.9%

Reading: Age, Heart_Rate and Last_Visit_Date are missing in about three quarters of rows, so the imputed median dominates those columns. Because missingness is this large, a **was_missing** flag is kept for each imputed column (section 3.3) and the visit date is reduced to Visit_Year. Raw Has_Disease had 2,520 blank and 2,506 “unknown” values out of 10,000; those 5,026 rows were removed first (10,000 → 4,974 rows, then 3 duplicates removed → 4,971).

Summary of filling missing values

Column type	Method	Fill value (training data)	Notes
Numeric (Age, BMI, Heart_Rate, Follow_Up, Cholesterol_Value, Systolic_BP, Diastolic_BP)	SimpleImputer(strategy='median')	Age 50, BMI 22.5, Heart_Rate 86, Follow_Up 14, Cholesterol 190, Systolic 120, Diastolic 80	Median is robust to outliers; fitted on train only
Visit_Year	SimpleImputer(strategy='median')	2022	
Categorical (Gender, Diabetic, Smoker, Medications, Cholesterol_Status)	SimpleImputer(strategy='most_frequent')	Mode of each column	Fitted on train only
Flags	_was_missing = isna() before imputing	0 / 1	11 flags kept; duplicate flags dropped

11. Visualization summary

Before preprocessing (EDA on cleaned data)

Visualization	What it shows
Histograms (Age, BMI, Heart_Rate, Follow_Up, Systolic_BP, Diastolic_BP, Cholesterol_Value)	Age and Heart_Rate are flat/uniform; BMI has a tall spike at 22.5; Systolic (120) and Diastolic (80) have a very tall single spike with thin tails; Follow_Up and Cholesterol_Value show only two bars (14/30 and 190/250).
Box plots	Age and Heart_Rate have no outliers; BMI shows many points above 27.5 and below 19.5; Follow_Up is a box spanning 14–30; Systolic and Diastolic collapse into a line at 120 / 80 with every other value drawn as an outlier (IQR = 0).
Count plots (Gender, City, Diabetic, Smoker, Cholesterol_Status/Level, Has_Disease, Blood_Pressure)	Categories are mostly balanced; New York is the largest city; Former is the largest smoker group; “120/80” dominates blood pressure; target is 50/50.
Date histogram and Visit_Year count plot	Visits between Jan 2020 and Oct 2025, spread evenly with no trend.
QQ plots (7 numeric variables)	Points do not follow the straight line: S-shaped for Age/Heart_Rate (uniform), step-shaped for Follow_Up/Cholesterol (two values), flat middle with curved ends for BMI and BP (spike at the median) → not normal.
Bivariate: numeric vs target (box plots)	Boxes for Has_Disease = 0 and 1 almost overlap for every numeric feature.
Bivariate: categorical vs target (count plots)	Bars for class 0 and 1 are nearly equal in every category.
Scatter plots (Age vs BMI, Age vs Heart_Rate, …; coloured by Has_Disease)	No pattern; horizontal line of points at BMI 22.5 (median spike); the two classes are mixed everywhere.
Correlation heatmap	All correlations within about ±0.06, including with Has_Disease.

After preprocessing (new notebook with plots)

Visualization	What to expect / what it confirms
Box plots after capping (real, non-missing values only)	Points beyond the 1st/99th-percentile bounds are clipped, so the count of values outside the bounds is 0 and the extreme dots disappear; Systolic/Diastolic are intentionally left uncapped to keep high and low BP.
Box plots before vs after (same scale)	Shows the tails removed from Age, BMI, Heart_Rate, Follow_Up and Cholesterol_Value. Plots are drawn on real values only; on full columns the imputed median spike (IQR ≈ 0) would make before/after look identical.
QQ plots after Yeo-Johnson	Points move closer to the line, especially for the skewed columns (BMI, Systolic), but remain stepped in the middle because 60–75% of values equal the imputed median. Data cannot become perfectly normal.
Skewness / kurtosis table before vs after	Printed by the notebook for the 7 transformed columns (Age, BMI, Heart_Rate, Follow_Up, Cholesterol_Value, Systolic_BP, Diastolic_BP).

12. Univariate, bivariate and multivariate analysis summary

Analysis	What was done	Result
Univariate – numerical	Mean, median, mode, quartiles, IQR, fences, min/max, skewness, kurtosis, variance, std; histograms, box plots, QQ plots	Age/Heart_Rate uniform; BMI and Systolic right-skewed; Follow_Up and Cholesterol are two-valued; BP spiked at 120/80.
Univariate – categorical	value_counts, percentage, mode, unique count; count plots	Mostly balanced classes; New York and Former smokers are the largest groups.
Univariate – date	Min/max date, histogram, Visit_Year count	2020-01-01 to 2025-10-01, even spread.
Bivariate – numeric vs target	Box plots and median by class	Medians nearly identical: Age 50 vs 48, BMI 22.5 vs 22.5, Heart_Rate 84 vs 86, Systolic 120 vs 120, Diastolic 80 vs 80, Cholesterol 190 vs 190; only Follow_Up median differs (14 vs 30).
Bivariate – categorical vs target	Count plots, row-percentage cross-tabs, chi-square test	Every category is about 50/50 (see tables below); all p-values > 0.05.
Bivariate – numeric vs numeric	Scatter plots, Age vs BMI by disease	No relationship.
Multivariate	Correlation matrix + heatmap, covariance matrix	All pairwise correlations weak (see section 13).

Chi-square test of each categorical variable against Has_Disease

Variable	Chi-square	p-value	Significant at 0.05?
Gender	0.452	0.5016	No
City	1.000	0.8011	No
Cholesterol_Level	0.858	0.8356	No
Diabetic	0.004	0.9520	No
Smoker	0.365	0.8331	No
Medications	0.627	0.4283	No
Diagnosis_Code	1.664	0.6450	No
Notes	1.082	0.5821	No
Cholesterol_Status	0.533	0.4653	No

Disease rate (class 1 %) by category

Variable	Class-1 percentage by category
Diabetic	No 50.4%; Yes 50.2%
Smoker	Former 49.5%; No 50.4%; Yes 49.0%
Gender	Female 49.8%; Male 48.5%
City	Boston 47.6%; Chicago 49.3%; Los Angeles 49.6%; New York 50.1%
Medications	Aspirin; Metformin 49.0%; Lisinopril; Metformin 50.6%
Diagnosis_Code	A00 48.8%; B20 51.4%; E11.9 49.3%; I10 48.6%

13. Correlation and covariance summary

Pair	Correlation	Covariance	Reading
Age – Cholesterol_Value	0.054	46.80	Largest positive pair, still negligible
Heart_Rate – Systolic_BP	–0.061	–13.04	Largest negative pair, negligible
Age – Diastolic_BP	–0.046	–7.93	Negligible
Age – BMI	0.024	2.82	None
Age – Heart_Rate	0.032	19.78	None
Follow_Up – Cholesterol_Value	–0.032	–7.69	None
Age – Has_Disease	–0.04	–0.557	Strongest feature–target link, still near 0
Heart_Rate – Has_Disease	0.04	0.396	Near 0
Other features – Has_Disease	≤ 0.02	≤ 0.06	Near 0

Reading: every correlation is between about –0.06 and +0.05, so the numeric features are practically independent of each other (no multicollinearity problem) and also independent of the target. Covariance values mostly reflect each variable's scale (e.g. Age variance 865, Heart_Rate 420), so correlation is the better measure of strength. Both correlation and covariance are computed on rows where both values are present.

14. Business / health decisions

Factor	Evidence in data	Decision
Age	Median 50 (no disease) vs 48 (disease)	Age alone does not separate patients; use as a feature and as age bands (Young/Adult/Middle_Age/Senior), not as a stand-alone rule.
BMI	Median 22.5 for both classes	BMI categories (Underweight … Obese) kept as a feature; no screening rule from BMI alone.
Heart rate	Median 84 vs 86	Small difference; keep in model, no threshold.
Systolic / Diastolic BP	Median 120 / 80 for both classes	Keep BP uncapped so genuine high/low BP is preserved for BP_Category and risk points.
Cholesterol	Median 190 for both classes	Use Cholesterol_Status (normal/high) and value as features; no strong signal.
Diabetic	50.4% vs 50.2% disease	No difference; do not target screening by diabetic status alone.
Smoker	Former 49.5%, No 50.4%, Yes 49.0%	No difference.
Follow_Up	Median 14 (no disease) vs 30 (disease)	The only visible difference; patients on a 30-day follow-up should be reviewed (but follow-up may be a result of the diagnosis – check for leakage before using it in practice).
Gender, City, Medications, Diagnosis code	All about 48–52%	No segment is riskier; services need not be targeted by these groups.
Overall	Correlations ≈ 0, chi-square p > 0.05, balanced classes	The current data cannot support a reliable risk model; collect more informative clinical data (labs, history, real BP variation) before deployment.

15. Outliers, transformation, feature engineering and encoding – summary

Topic	What was done	Settings / reason
Outliers (detection)	1.5×IQR rule, box plots	IQR = 0 for BP and ≈ 0 for BMI because most values equal the median, so the IQR rule flags almost everything; therefore percentile capping is used.
Outliers (treatment)	Percentile capping (winsorizing) with apply_caps()	Columns: Age, BMI, Heart_Rate, Follow_Up, Cholesterol_Value. Lower/upper quantiles 0.01 / 0.99, learned from real (non-missing) training values only, applied with clip() to train and test. Systolic_BP and Diastolic_BP are deliberately not capped to keep high and low BP.
Transformation	PowerTransformer(method='yeo-johnson', standardize=False), one per column	Columns: Age, BMI, Heart_Rate, Follow_Up, Cholesterol_Value, Systolic_BP, Diastolic_BP. Fitted on train only; applied after feature engineering because bins and risk points need real units (years, kg/m², mmHg). Reduces skew; cannot give a perfect normal shape because of the spike at the median.
Feature engineering	Medication_Count, Age_Group, BMI_Category, BP_Category, Risk_Points, Risk_Score, Visit_Year, *_was_missing flags	Bins: Age [0,18,40,60,100]; BMI [0,18.5,25,30,100]; Risk_Points [-1,2,5,100] → Low/Medium/High. Dropped: Patient_ID, Name, Blood_Pressure (split), Cholesterol_Level (split), Last_Visit_Date (→ Visit_Year), Diagnosis_Code, Notes, City.
Encoding	OneHotEncoder and OrdinalEncoder	One-hot (drop='first', handle_unknown='ignore', sparse_output=False): Gender, Diabetic, Smoker, Medications, Cholesterol_Status. Ordinal (explicit order, unknown_value=-1): Age_Group, BMI_Category, Risk_Score.
Feature selection	SelectKBest(score_func=f_classif)	k = 6 with Risk_Score forced in: the other 5 are the highest F-scores. Scores are ranked with k='all', then the chosen columns are kept in training order.
Scaling	StandardScaler	Fitted on the selected training features only.

16. Hyperparameters and metrics used – overview

Stage	Parameters
Data split	train_test_split(test_size=0.20, random_state=0)
Imputation	Median (numeric, Visit_Year) / most_frequent (categorical)
Capping	LOWER_Q = 0.01, UPPER_Q = 0.99
Transformation	PowerTransformer(method='yeo-johnson', standardize=False)
Encoding	OneHotEncoder(drop='first', handle_unknown='ignore', sparse_output=False); OrdinalEncoder(unknown_value=-1)
Selection / scaling	SelectKBest(f_classif, k=6, forced Risk_Score); StandardScaler()
Baseline models	Logistic Regression (max_iter=1000), SVM linear/RBF, KNN (n_neighbors=6, p=2), GaussianNB, Decision Tree (entropy), Random Forest (n_estimators=10, entropy)
Tuning	GridSearchCV(cv=5, scoring='f1_weighted', refit=True, n_jobs=-1, verbose=3); grids and best parameters in section 4.2
Final saved model	best_models['Random Forest'] (tuned) in RandomForest_model.pkl

Metric	Use
Accuracy (accuracy_score)	Share of correct predictions (balanced target, so meaningful)
Weighted F1 (f1_score, average='weighted')	Tuning score in GridSearchCV; balance of precision and recall
ROC-AUC (roc_auc_score)	Ability to separate classes; 0.5 = random
Confusion matrix	TN, FP, FN, TP counts for each model

Metric	Use
Classification report	Precision, recall, F1 per class
Success criteria	Recall ≥ 80%, F1 ≥ 80%, ROC-AUC ≥ 0.80 (not met: scores ≈ 0.5, see section 5)